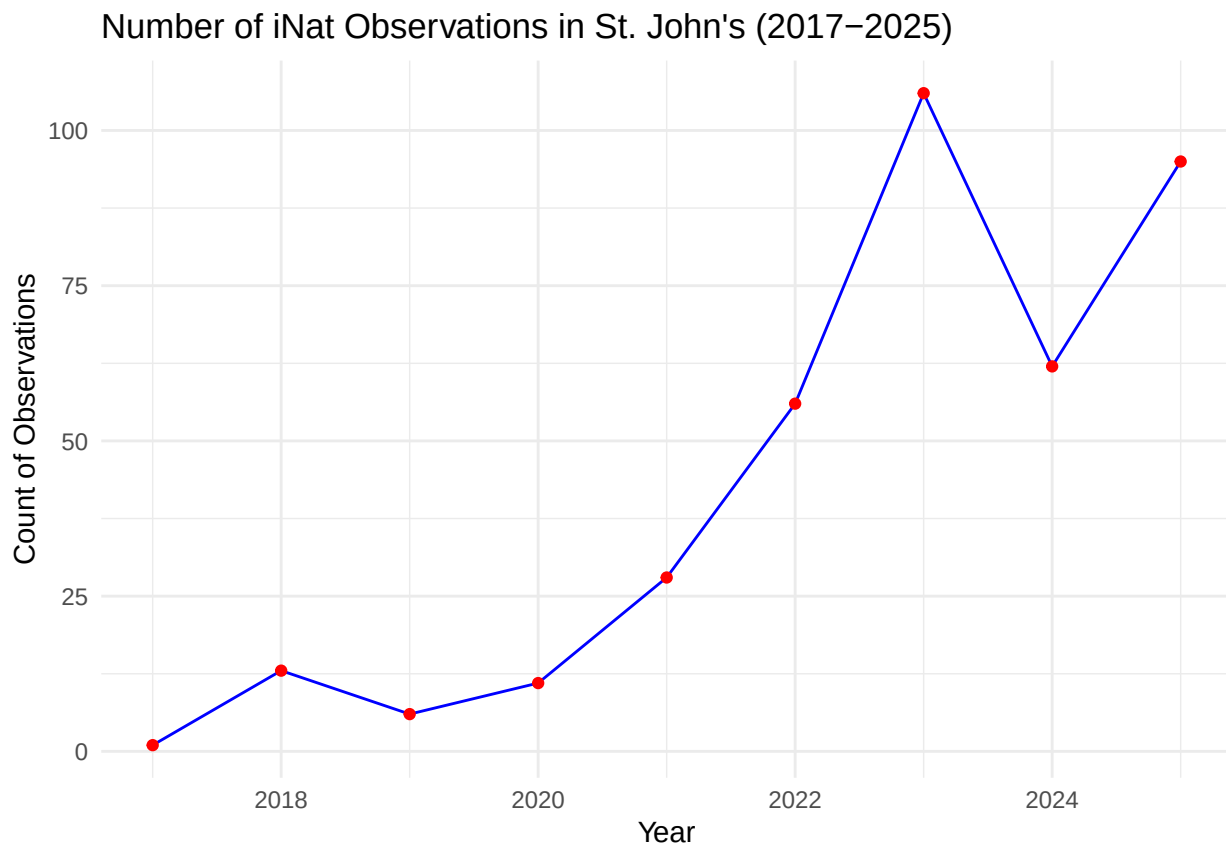


St. John's Citizen Science Analysis

Audrey Yang, Archith Sharma, Jenny Lyu

iNat in St. John's had much more activity than eBird did, but aggregate iNat data isn't available. So, we had to use the iNat API to get the data for St. John's and then calculated the growth rate from 2021 to 2025.

Since aggregate iNat data isn't available, we can't directly compare the growth rate of iNat observations in St. John's to Pennsylvania as a whole, but we can use eBird data for Pennsylvania as a baseline to compare the growth rate of iNat observations in St. John's to the growth rate of eBird checklists, a separate, but equally valid proxy.



The growth rate of iNat observations in St. John's is much higher (**4.8 times**) than the growth rate of eBird checklists in Pennsylvania, which suggests that **engagement in that area is growing much faster than in the overall state.**

The comparison to this baseline also suggests that the growth in iNat observations in St. John's is **not just part of a general trend** of increasing participation in citizen science, but is **likely influenced by ALT engagement** in that area, while it is important to note that correlation doesn't imply causation.

In addition, eBird in the Brighton Heights neighborhood of Pittsburgh, PA, has seen **3.74 times** YoY growth compared to eBird PA since the ALT project started. This also suggests that the ALT project may have had

a significant impact on increasing engagement in citizen science in that area, and that the growth in eBird checklists in Brighton Heights is likely influenced by the ALT project as well.

The trade-off is that **while eBird data is slightly weaker and not as localized to St. John's, the comparison is to the same source of data, so we can be more confident that the growth rate difference is due to the ALT project rather than differences in data collection methods or other factors.** iNat's data is much more localized to St. John's, but the comparison is to a different source of data, so we have to be more cautious. **Using both these datasets together allows us to be more confident in our conclusions about the impact of the ALT project** on increasing citizen science engagement in St. John's.

Both of these datasets **in conjunction suggest that the ALT project has had a significant impact on increasing engagement in citizen science through the St. John's project**, and that this impact is reflected in the growth rates of both iNat observations and eBird checklists in that area compared to the overall state of Pennsylvania.

```
## Annual growth rate of iNat observations in St. John's
## from 2021 to 2025: 35.72 %

## Annual growth rate of eBird observations in Brighton Heights
## from 2021 to 2025: 27.8 %

## Annual growth rate of eBird checklists in PA
## from 2021 to 2025: 7.44 %

## Factor difference in growth rates
## (iNat St. John's vs eBird Allegheny): 4.8

## Factor difference in growth rates
## (eBird Brighton Heights vs eBird PA County): 3.74
```

Another comparison would be to Allegheny County's eBird data. In this analysis, **the iNat/eBird St. John's/Brighton Heights growth rate is still much higher than the eBird Allegheny County growth rate, but the difference is slightly smaller** (2.84 times faster growth for iNat St. John's and 2.21 times for Brighton Heights eBird compared to eBird Allegheny County). All the same conclusions about the impact of the ALT project on increasing engagement in citizen science in St. John's and Brighton Heights still hold, but **the comparison to Allegheny County's eBird data provides a more localized baseline for comparison**, which strengthens our confidence that the growth rate differences are due to the ALT project rather than other factors.

```
## Annual growth rate of iNat observations in St. John's
## from 2021 to 2025: 35.72 %

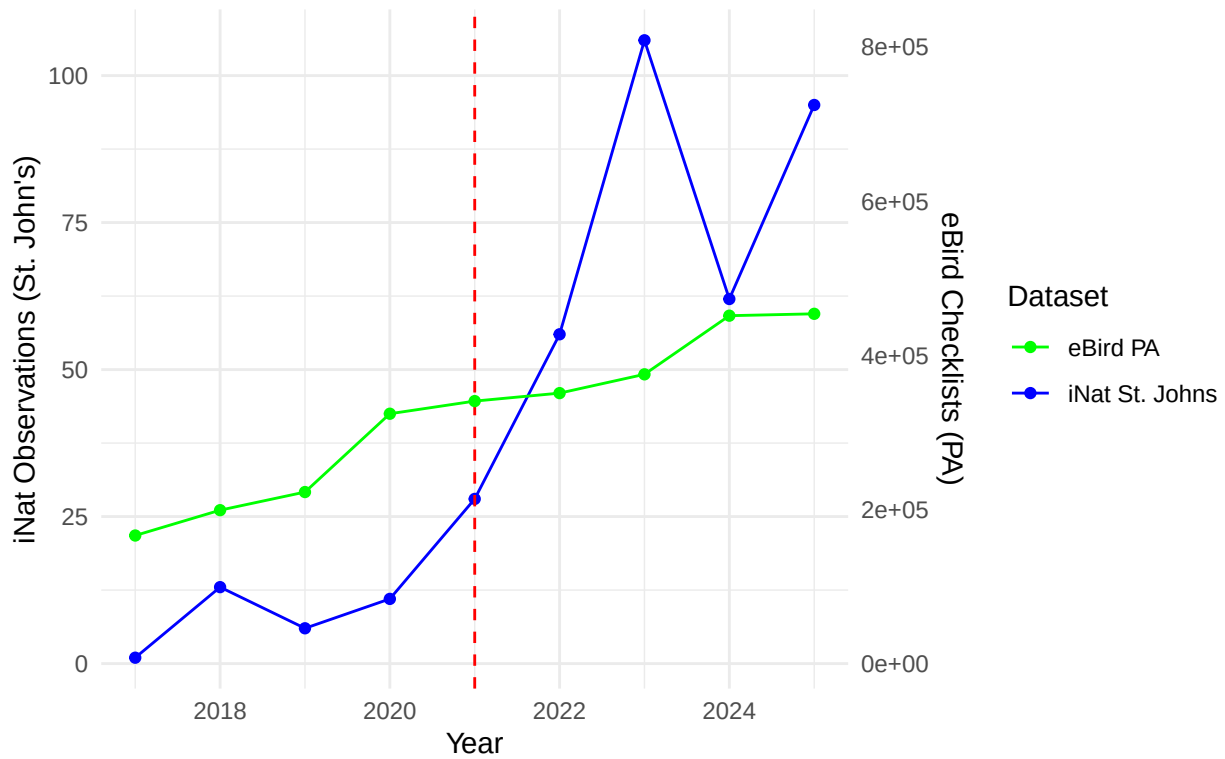
## Annual growth rate of eBird observations in Brighton Heights
## from 2021 to 2025: 27.8 %

## Annual growth rate of eBird checklists in Allegheny County
## from 2021 to 2025: 12.57 %

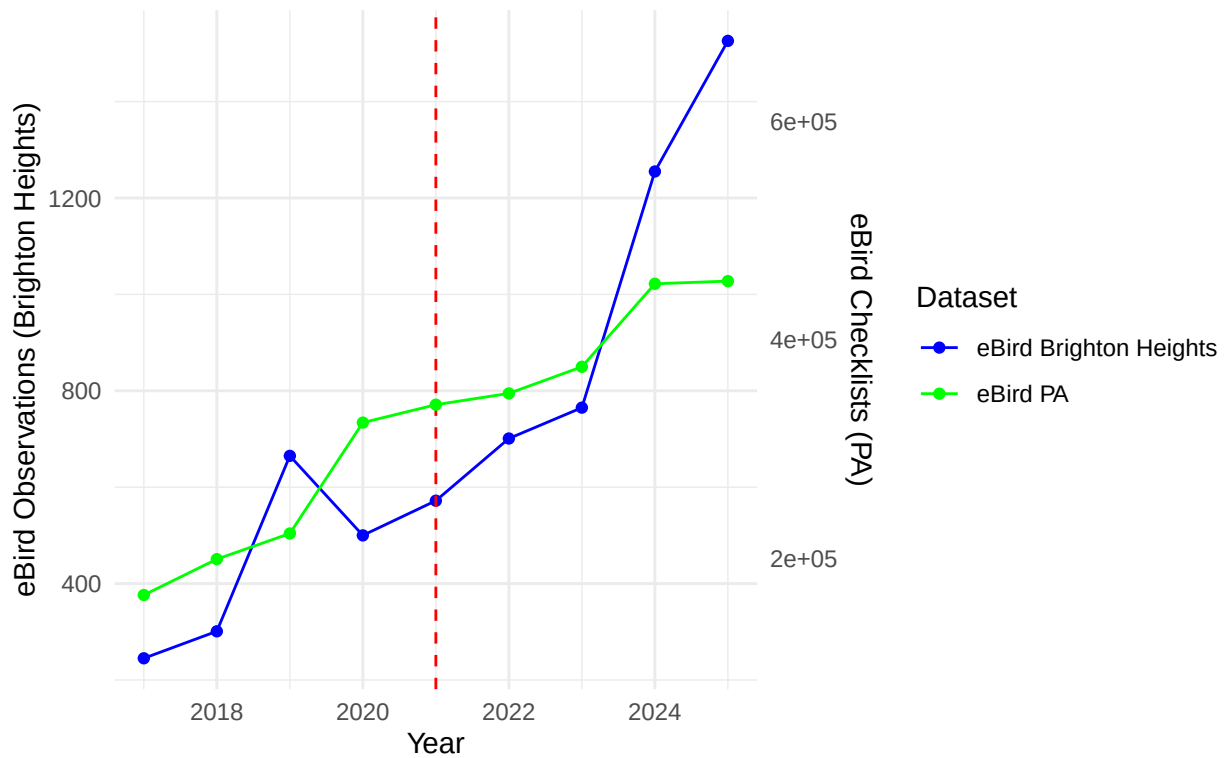
## Factor difference in growth rates
## (iNat St. John's vs eBird Allegheny): 2.84

## Factor difference in growth rates
## (eBird Brighton Heights vs eBird Allegheny County): 2.21
```

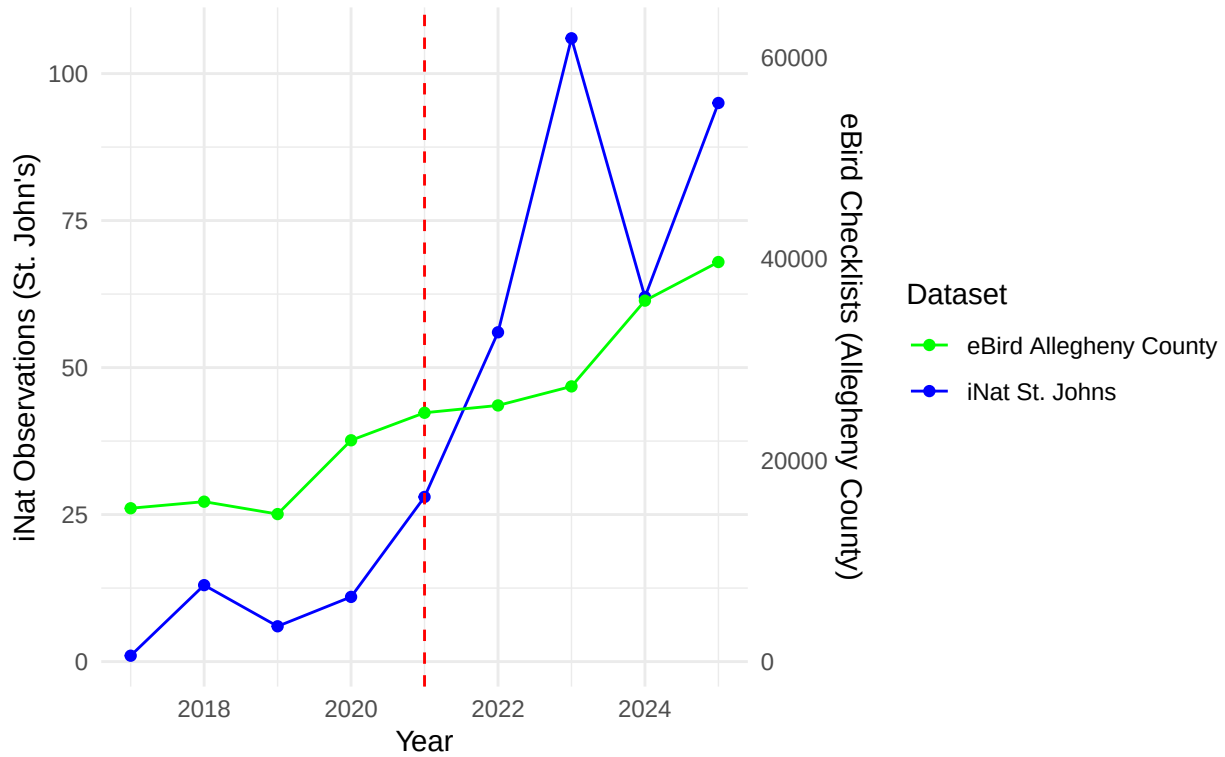
Growth Comparison:
iNat St. John's vs. eBird PA (2017–2025)



Growth Comparison:
eBird Brighton Heights vs. eBird PA (2017–2025)



Growth Comparison:
iNat St. John's vs. eBird Allegheny County (2017–2025)



Growth Comparison:
eBird Brighton Heights vs. eBird Allegheny County (2017–2025)

